

HESI PREP TEST

Explanation of HESI scores:

90% to 100%: **Excellent**

Super job! You have excellent Math skills and should have no difficulty calculating medication administration problems in your program.

80% to 89%: **Very Good**

Good job! You have very good Math skills and should be able to calculate medication administration problems in your program. However, you may need a little extra assistance from a knowledgeable person when preparing for medication administration tests.

79% to 75%: **Satisfactory**

You have adequate Math skills but are likely to need additional outside tutoring when learning to calculate medication administration problems.

74% or below: **Needs Improvement**

You need to review and practice Math problems.

You would likely benefit from enrolling in an adult Math remediation program prior to entering your program or in conjunction with the beginning level courses.

Subject Area	Number of questions
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Math	50
Decimals	11
Fractions	23
Household measures	2
General Math	9
Algebra	10

HESI MATH PRACTICE PROBLEM

1. **Divide and simplify:** $6 \div 2\frac{2}{3}$
 - a. $2\frac{2}{8}$
 - b. 16
 - c. $2\frac{1}{4}$
 - d. $12\frac{2}{3}$

2. **Convert 0.60 to ratio in simplest form.**
 - a. 2:50
 - b. 2:6
 - c. 1:6
 - d. 3:5

3. **A shoe company sends out 7 trucks loaded with nursing shoes that cost the company \$1,572 per truck. If they send out 750 boxes per truck, what would be the profit if the cost per box is \$ 125.40?**
 - a. 79,490.40
 - b. 614,400.00
 - c. 45,357.60
 - d. 647,346.00

4. **Calculate the quotient of $\frac{54,256}{45}$ (Round to the nearest tenth)**

5. **Multiply and reduced to lowest terms** $\frac{5}{13} \times 3\frac{3}{8}$
 - a. 45/104
 - b. 104/45
 - c. $1\frac{31}{104}$
 - d. 5/9

6. **Add:** $5\frac{2}{3} + \frac{4}{7}$
 - a. 21/131
 - b. $6\frac{5}{21}$
 - c. $1\frac{2}{5}$
 - d. $5\frac{26}{21}$

7. The average speed of a high velocity train is 200 kilometers per hour. The distance between Paris and Naples is 7,346 km (1 mile = 1.609 km). How long will the trip take? (Round to the nearest whole number).
- a. 38 hours
 - b. 36.73 hours
 - c. 37 hours
 - d. 3.8 hours
8. How many grams in 450 milligrams?
- a. 45 grams
 - b. 0.45 grams
 - c. 4.5 grams
 - d. 450 grams
9. $23/15$ divided by $16/17$ =
- a. $391/240$
 - b. $3\frac{3}{4}$
 - c. $1\frac{151}{240}$
 - d. 1
10. Convert 25.52 to a reduced fraction
- a. $25/52$
 - b. $638/52$
 - c. $2552/100$
 - d. $25\frac{13}{25}$
11. Convert $3/5$ to a decimal.
- a. 0.06
 - b. 1.6
 - c. 0.6
 - d. 3.50
12. Add $6\frac{1}{4} + 8\frac{1}{3} =$
- a. $14\frac{2}{7}$
 - b. $\frac{19}{4}$
 - c. $175/12$
 - d. $14\frac{7}{12}$
13. A recent online survey published that a hospital needs 9 nurses for every 45 patients. How many nurses a hospital would need in order to take care of 255 patients?
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14. Convert $\frac{5}{8}$ to a decimal

- a. 0.0625
- b. 6.25
- c. 0.625
- d. 1.6

15. Convert 0.65 to a fraction in lowest term

- a. $\frac{1}{65}$
- b. $\frac{20}{13}$
- c. $\frac{13}{20}$
- d. $\frac{65}{100}$

16. A book store contains 75,972 books, divided in three sections: Science, Business, and Math. The science section represents $\frac{1}{3}$ of the total books. Of the remaining books, 75% are business. How many books are in the math section?

- a. 50,648
- b. 37986
- c. 12,662
- d. 25324

17. During final exams 172 students visited the math lab in total. If 4 out of 11 were male what is the percentage of female students who visited the math lab? (Round to the nearest tenth).

- a. 63.6%
- b. 63.64%
- c. 62.5%
- d. 63%

18. 0.2 divided by 1.1?

- a. 1.818
- b. 0.1818
- c. 181.8
- d. 18.18

19. What is 5 percent of 50?

20. Solve for x: $44:68::x:236$ (Round to the nearest tenth).

- a. $x=155.53$
- b. $x=152.7$
- c. $x=88.65$
- d. $x=135$

21. Miguel wants to get cheaper gas, so he drives out of his way to buy gas at \$2.50 instead of \$2.75 at his local station. If he needs 15.5 gallons to fill his SUV, how much did he save by going to the less expensive station? (Round to the nearest hundredth).

- a. \$3.875
- b. \$3.87
- c. \$3.88
- d. \$3.00

22. Ben walks 25 miles from home to work. If he walks the first 5 miles in 1 hour, the second 5 miles in 1.4 hours, the third 5 miles in 1.7 hours and the last 5 miles in 1.8 hours, how long will it take him to complete the 25 mile walking distance? (Enter numeric value only. If rounding is required, round to the nearest tenth).

23. Convert 250 milliliters to liters

- a. 2.5L
- b. 25L
- c. 0.25L
- d. 250000L

24. Convert the metric equivalent: 500 mg equals x grams. (Enter the numerical value only).

25. How many liters are in 300 milliliters?

- a. 30 liters
- b. 0.3 liters
- c. 3 liters
- d. 0.03 liters

26. Ratio and proportion: $0.5:20::x:200$

- a. $x = 25$
- b. $x = 5$
- c. $x = 50$
- d. $x = 4$

27. Change $66/4$ to a decimal

- a. 1.65
- b. $32/2$
- c. 0.165
- d. 16.5

28. Which is the largest decimal?

- a. 0.0688
- b. 0.667
- c. 0.68
- d. 0.6

29. Convert the metric equivalent: 8,500 grams equals x kg. (Enter numerical value only).

30. Convert the fraction into decimals $7/4=175\%$ (Enter numerical value only).

31. How many milliliters are contained in 3 table spoons of fluid? (Enter numerical value only).

32. Solve for x: 5:50:: x:500

- a. 4
- b. 50
- c. 5
- d. 5.5

33. Convert $2/3$ to a decimal

- a. 0.2222(repeating)
- b. 0.3333 (repeating)
- c. 0.6666 (repeating)
- d. 1.5

34. Subtract $5\frac{2}{3} - 2\frac{6}{7}$ (Enter numerical value only).

35. How many pints are in 80 ounces?

- a. 16 pints
- b. 1,280 pints
- c. 15 pints
- d. 5 pints

36. There are 200 ounces in 6 liters. How many milliliters (ml) are in 6 liters? (Enter numerical value only).

37. How many gallons are in 576 ounces?

- a. 4.5 gallons
- b. 288 gallons
- c. 4 gallons
- d. 5 gallons

38. How many ounces are in $8\frac{1}{4}$ pints?

- a. 136 oz
- b. 132 oz
- c. 2.25 oz
- d. 128 oz

39. Multiply 1.25×2.0

- a. 250.0
- b. 2.50
- c. 0.250
- d. 25.0

40. Add $\frac{4}{5} + \frac{3}{15}$ (reduced to lowest terms)

- a. 1
- b. $\frac{7}{15}$
- c. $\frac{12}{15}$
- d. $\frac{1}{15}$

41. How far does a runner run (in kilometers) in a 1,000 meter race?

- a. 1 km
- b. $\frac{1}{2}$ km
- c. 3 km
- d. 0.10 km

42. Henry was able to help 4 out of 9 of students that requested help in the math lab and Michelle helped 45% of the total students who asked for help. Who helped more students, Henry or Michelle?

- a. The amount cannot be calculated.
- b. Henry
- c. Michelle
- d. They both helped the same number of students.

43. 27 students attended a HESI session in the Math Lab if 8 out of 11 were females, how many were male? (Round to a whole number)

44. 3 gallons is equal to how many ounces?

45. $5\frac{1}{2}$ divide by $2\frac{1}{2}$

- a. $2\frac{1}{2}$
- b. 2.0
- c. $2\frac{1}{5}$
- d. $2\frac{3}{4}$

46. How many teaspoons are there in 5 tablespoons?

- a. 1.67 Tsp.
- b. 0.6 Tsp.
- c. 15 Tsp.
- d. 1.5 Tsp.

47. Evaluate the expression: $4x - 5y$ $x = -3$ and $y = 2$

- a. 22
- b. -2
- c. 2
- d. -22

48. Solve the equation for m: $2m + 1 = -3m - 1$

- a. $\frac{2}{5}$
- b. $-\frac{5}{2}$
- c. $-\frac{2}{5}$
- d. 0

49. Dr. Henry has his nursing staff line up at 4:30PM for the annual meeting. What would that be in military time?

- a. 0430
- b. 4030
- c. 1630
- d. 2030

50. 330 is 165% of what number

- a. 150
- b. 200
- c. 20
- d. 545

51. Write 1,763 as a Roman numeral

- a. MDCCLII
- b. MCDXXXVI
- c. MDCCLXIII
- d. MDCCLXII

52. What is the regular time for the given military time: 0507?

- a. 5:07 A.M
- b. 5:07 P.M
- c. 7:05 A.M
- d. 7:07 P.M

53. Subtract: $3\frac{1}{2} - 2\frac{2}{3}$

- a. $\frac{6}{5}$
- b. $1\frac{1}{5}$
- c. $\frac{5}{6}$
- d. $\frac{5}{16}$

54. What is the correct answer for MDCXLIV?

- a. 1544
- b. 1654
- c. 1554
- d. 1644

55. What % of 200 is 34?

- a. 34%
- b. 68%
- c. 17%
- d. 20%.

Answer Key

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|---------------|-----------------------|----------------|
| 1. C | 22. 5.9 hours | 43. 7 males |
| 2. D | 23. C | 44. 384 ounces |
| 3. D | 24. 0.5 grams | 45. C |
| 4. 1,205.7 | 25. B | 46. C |
| 5. C | 26. B | 47. D |
| 6. B | 27. D | 48. C |
| 7. C | 28. C | 49. C |
| 8. B | 29. 8.5 kilograms | 50. B |
| 9. C | 30. 1.75 | 51. C |
| 10. D | 31. 45 milliliters | 52. A |
| 11. C | 32. B | 53. C |
| 12. D | 33. C | 54. D |
| 13. 51 nurses | 34. $2\frac{17}{21}$ | 55. C |
| 14. C | 35. D | |
| 15. C | 36. 6,000 milliliters | |
| 16. C | 37. A | |
| 17. A | 38. B | |
| 18. B | 39. B | |
| 19. 2.5 | 40. A | |
| 20. B | 41. A | |
| 21. C | 42. C | |

